Limit to 1000 rows

```
# Project: SmartMart Retail Sales Analysis
# Name: Shubham Kumar
# Course: B.tech CSE (D.A)
# Description: Practice assignment covering select, where, group by,
#joins and business queries

# create database
create database smartmart_db;

# use database
use smartmart_db;

# create customers table
create table customers (
    customer_id int primary key,
    customer_name varchar(50),
    city varchar(50),
    age int
);

# create products table
```

Output

Action Output

#	Time	Action	Message
1	14:15:51	create database smartmart_db	1 row(s) affected
2	14:15:56	use smartmart_db	0 row(s) affected
3	14:16:01	create table customers (customer_id int primary key, customer_name varchar(50), city varchar(50), age int)	0 row(s) affected

Limit to 1000 rows

```
# create products table
create table products (
    product_id int primary key,
    product_name varchar(100),
    category varchar(50),
    price decimal(10,2)
);

# create sales table
create table sales (
    order_id int primary key,
    customer_id int,
    product_id int,
    quantity int,
    order_date date
);

# insert customers
insert into customers values
(1,'Rahul Sharma','Delhi',28),
```

Output

Action Output

#	Time	Action	Message
1	14:15:51	create database smartmart_db	1 row(s) affected
2	14:15:56	use smartmart_db	0 row(s) affected
3	14:16:01	create table customers (customer_id int primary key, customer_name varchar(50), city varchar(50), age int)	0 row(s) affected
4	14:16:46	create table products (product_id int primary key, product_name varchar(100), category varchar(50), price ...	0 row(s) affected
5	14:16:50	create table sales (order_id int primary key, customer_id int, product_id int, quantity int, order_date date)	0 row(s) affected

Limit to 1000 rows

```sql
insert into customers values
(1,'Rahul Sharma','Delhi',28),
(2,'Priya Verma','Mumbai',32),
(3,'Amit Singh','Bangalore',25),
(4,'Neha Kapoor','Delhi',29),
(5,'Rohit Mehta','Pune',35),
(6,'Anjali Gupta','Mumbai',27),
(7,'Karan Patel','Ahmedabad',31),
(8,'Sneha Reddy','Hyderabad',26);

# insert products
insert into products values
(101,'Laptop','Electronics',80000),
(102,'Smartphone','Electronics',40000),
(103,'Headphones','Electronics',2000),
(104,'T-Shirt','Clothing',800),
(105,'Jeans','Clothing',2000),
(106,'Microwave','Home Appliances',7000),
(107,'Refrigerator','Home Appliances',30000),
(108,'Air Conditioner','Home Appliances',45000);
```

Output

Action Output

#	Time	Action	Message
3	14:16:01	create table customers (customer_id int primary key, customer_name varchar(50), city varchar(50), age int)	0 row(s) affected
4	14:16:46	create table products (product_id int primary key, product_name varchar(100), category varchar(50), pric...	0 row(s) affected
5	14:16:50	create table sales (order_id int primary key, customer_id int, product_id int, quantity int, order_date date)	0 row(s) affected
6	14:17:21	insert into customers values (1,'Rahul Sharma','Delhi',28), (2,'Priya Verma','Mumbai',32), (3,'Amit Singh','Bangalore',2...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0
7	14:17:26	insert into products values (101,'Laptop','Electronics',80000), (102,'Smartphone','Electronics',40000), (103,'Headph...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0

Limit to 1000 rows

```
# insert sales
insert into sales values
(1001,1,101,1,'2024-01-10'),
(1002,2,102,2,'2024-01-11'),
(1003,3,104,3,'2024-01-12'),
(1004,4,105,2,'2024-01-12'),
(1005,5,103,4,'2024-01-13'),
(1006,6,106,1,'2024-01-14'),
(1007,7,108,1,'2024-01-14'),
(1008,8,102,1,'2024-01-15'),
(1009,1,103,2,'2024-01-16'),
(1010,2,104,5,'2024-01-17'),
(1011,3,105,1,'2024-01-18'),
(1012,4,101,1,'2024-01-19');

-----------------------------------------------------

-----------------------------------------------------
# Level 1 - Basic SELECT

# Q1. display all customers
```

Output

Action Output

#	Time	Action	Message
4	14:16:46	create table products (product_id int primary key, product_name varchar(100), category varchar(50), pric...	0 row(s) affected
5	14:16:50	create table sales (order_id int primary key, customer_id int, product_id int, quantity int, order_date date)	0 row(s) affected
6	14:17:21	insert into customers values (1,'Rahul Sharma','Delhi',28), (2,'Priya Verma','Mumbai',32), (3,'Amit Singh','Bangalore',2...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0
7	14:17:26	insert into products values (101,'Laptop','Electronics',80000), (102,'Smartphone','Electronics',40000), (103,'Headph...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0
8	14:18:02	insert into sales values (1001,1,101,1,'2024-01-10'), (1002,2,102,2,'2024-01-11'), (1003,3,104,3,'2024-01-12'), (100...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0

Limit to 1000 rows

```
# Level 1 - Basic SELECT

# Q1. display all customers
select *
from customers;

# Q2. show only customer_name and city
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

customer_id	customer_name	city	age
1	Rahul Sharma	Delhi	28
2	Priya Verma	Mumbai	32
3	Amit Singh	Bangalore	25
4	Neha Kapoor	Delhi	29
5	Rohit Mehta	Pune	35
6	Anjali Gupta	Mumbai	27
7	Karan Patel	Ahmedabad	31
8	Sneha Reddy	Hyderabad	26
NULL	NULL	NULL	NULL

Limit to 1000 rows

```
# Q2. show only customer_name and city
select customer_name, city
from customers;

# Q3. display all products in electronics category
select *
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_name	city
Rahul Sharma	Delhi
Priya Verma	Mumbai
Amit Singh	Bangalore
Neha Kapoor	Delhi
Rohit Mehta	Pune
Anjali Gupta	Mumbai
Karan Patel	Ahmedabad
Sneha Reddy	Hyderabad

Limit to 1000 rows

```
from customers;

# Q3. display all products in electronics category
select *
from products
where category = 'Electronics';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	price
101	Laptop	Electronics	80000.00
102	Smartphone	Electronics	40000.00
103	Headphones	Electronics	2000.00
NULL	NULL	NULL	NULL

Limit to 1000 rows

```
# Q4. show products with price greater than 10000
select *
from products
where price > 10000;

# Q5. display unique categories
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	price
101	Laptop	Electronics	80000.00
102	Smartphone	Electronics	40000.00
107	Refrigerator	Home Appliances	30000.00
108	Air Conditioner	Home Appliances	45000.00
NULL	NULL	NULL	NULL

Limit to 1000 rows

```
# Q5. display unique categories
select distinct category
from products;

-----------------------------------------------------
# Level 2 - WHERE Conditions
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category
Electronics
Clothing
Home Appliances

Limit to 1000 rows

```
102 ----------------------------------------------------
103 # Level 2 - WHERE Conditions
104
105 # Q6. show customers who live in delhi
106 select *
107 from customers
108 where city = 'Delhi';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

customer_id	customer_name	city	age
1	Rahul Sharma	Delhi	28
4	Neha Kapoor	Delhi	29
NULL	NULL	NULL	NULL

Limit to 1000 rows

```
# Q7. show customers with age greater than 30
select *
from customers
where age > 30;

# Q8. display products with price between 2000 and 50000
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

customer_id	customer_name	city	age
2	Priya Verma	Mumbai	32
5	Rohit Mehta	Pune	35
7	Karan Patel	Ahmedabad	31
NULL	NULL	NULL	NULL

Limit to 1000 rows

```
# Q8. display products with price between 2000 and 50000
select *
from products
where price between 2000 and 50000;

# Q9. find orders placed after 2024-01-12
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	price
102	Smartphone	Electronics	40000.00
103	Headphones	Electronics	2000.00
105	Jeans	Clothing	2000.00
106	Microwave	Home Appliances	7000.00
107	Refrigerator	Home Appliances	30000.00
108	Air Conditioner	Home Appliances	45000.00
NULL	NULL	NULL	NULL

Limit to 1000 rows

```
# Q9. find orders placed after 2024-01-12
select *
from sales
where order_date > '2024-01-12';

# Q10. display sales where quantity > 2
select *
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

order_id	customer_id	product_id	quantity	order_date
1005	5	103	4	2024-01-13
1006	6	106	1	2024-01-14
1007	7	108	1	2024-01-14
1008	8	102	1	2024-01-15
1009	1	103	2	2024-01-16
1010	2	104	5	2024-01-17
1011	3	105	1	2024-01-18
1012	4	101	1	2024-01-19
NULL	NULL	NULL	NULL	NULL

Limit to 1000 rows

```
# Q10. display sales where quantity > 2
select *
from sales
where quantity > 2;

-----------------------------------------------------
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

order_id	customer_id	product_id	quantity	order_date
1005	5	103	4	2024-01-13
1006	6	106	1	2024-01-14
1007	7	108	1	2024-01-14
1008	8	102	1	2024-01-15
1009	1	103	2	2024-01-16
1010	2	104	5	2024-01-17
1011	3	105	1	2024-01-18
1012	4	101	1	2024-01-19
NULL	NULL	NULL	NULL	NULL

Limit to 1000 rows

```
-----------------------------------------------------
# Level 3 - Sorting & Limiting

# Q11. display products sorted by price (highest first)
select *
from products
order by price desc;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	price
101	Laptop	Electronics	80000.00
108	Air Conditioner	Home Appliances	45000.00
102	Smartphone	Electronics	40000.00
107	Refrigerator	Home Appliances	30000.00
106	Microwave	Home Appliances	7000.00
103	Headphones	Electronics	2000.00
105	Jeans	Clothing	2000.00
104	T-Shirt	Clothing	800.00
NULL	NULL	NULL	NULL

Limit to 1000 rows

```
# Q12. show top 3 most expensive products
select *
from products
order by price desc limit 3;

# Q13. display customers sorted by age ascending
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

product_id	product_name	category	price
101	Laptop	Electronics	80000.00
108	Air Conditioner	Home Appliances	45000.00
102	Smartphone	Electronics	40000.00
NULL	NULL	NULL	NULL

Limit to 1000 rows

```
# Q13. display customers sorted by age ascending
select *
from customers
order by age;

# Q14. show latest 5 orders
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

customer_id	customer_name	city	age
3	Amit Singh	Bangalore	25
8	Sneha Reddy	Hyderabad	26
6	Anjali Gupta	Mumbai	27
1	Rahul Sharma	Delhi	28
4	Neha Kapoor	Delhi	29
7	Karan Patel	Ahmedabad	31
2	Priya Verma	Mumbai	32
5	Rohit Mehta	Pune	35
NULL	NULL	NULL	NULL

Limit to 1000 rows

```
# Q14. show latest 5 orders
select *
from sales
order by order_date desc limit 5;

-----------------------------------------------------
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

order_id	customer_id	product_id	quantity	order_date
1012	4	101	1	2024-01-19
1011	3	105	1	2024-01-18
1010	2	104	5	2024-01-17
1009	1	103	2	2024-01-16
1008	8	102	1	2024-01-15
NULL	NULL	NULL	NULL	NULL

Limit to 1000 rows

```
153     ----------------------------------------------------
154     # Level 4 - NULL Handling
155
156     # Update a row to create null value
157     update customers
158     set city = null
159     where customer_id = 8;
160
```

Limit to 1000 rows

```
# Q15. find customers where city is null
select *
from customers
where city is null;

# Q16. find customers where city is not null
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

customer_id	customer_name	city	age
8	Sneha Reddy	NULL	26
NULL	NULL	NULL	NULL

Limit to 1000 rows

```
# Q16. find customers where city is not null
select *
from customers
where city is not null;

-----------------------------------------------------
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

customer_id	customer_name	city	age
1	Rahul Sharma	Delhi	28
2	Priya Verma	Mumbai	32
3	Amit Singh	Bangalore	25
4	Neha Kapoor	Delhi	29
5	Rohit Mehta	Pune	35
6	Anjali Gupta	Mumbai	27
7	Karan Patel	Ahmedabad	31
NULL	NULL	NULL	NULL

Limit to 1000 rows

```
-----------------------------------------------------
# Level 5 - Aggregations

# Q17. find total number of customers
select count(*) as total_customers
from customers;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_customers
8

Limit to 1000 rows

```
# Q18. find average product price
select avg(price) as avg_price
from products;

# Q19. find maximum product price
select max(price) as max_price
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

avg_price
25850.000000

Limit to 1000 rows

```
# Q19. find maximum product price
select max(price) as max_price
from products;

# Q20. find total quantity sold
select sum(quantity) as total_quantity
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

max_price
80000.00

Limit to 1000 rows

```
# Q20. find total quantity sold
select sum(quantity) as total_quantity
from sales;

-----------------------------------------------------
# Level 6 - GROUP BY
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_quantity
24

Limit to 1000 rows

```
190     ----------------------------------------------------
191     # Level 6 - GROUP BY
192
193     # Q21. find number of products in each category
194     select category, count(*) as total_products
195     from products
196     group by category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	total_products
Electronics	3
Clothing	2
Home Appliances	3

Limit to 1000 rows

```
# Q22. find total sales quantity by product_id
select product_id, sum(quantity) as total_quantity
from sales
group by product_id;

# Q23. find average price by category
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_id	total_quantity
101	2
102	3
104	8
105	3
103	6
106	1
108	1

Limit to 1000 rows

```
# Q23. find average price by category
select category, avg(price) as avg_price
from products
group by category;

-----------------------------------------------------
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	avg_price
Electronics	40666.666667
Clothing	1400.000000
Home Appliances	27333.333333

Limit to 1000 rows

```
# Level 7 - HAVING Clause

# Q24. find categories where average price > 10000
select category, avg(price)
from products
group by category
having avg(price) > 10000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	avg(price)
Electronics	40666.666667
Home Appliances	27333.333333

Limit to 1000 rows

```
# Q25. find products where total quantity sold > 3
select product_id, sum(quantity)
from sales
group by product_id
having sum(quantity) > 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_id	sum(quantity)
104	8
103	6

Limit to 1000 rows

```
# Q26. find categories having more than 2 products
select category, count(*)
from products
group by category
having count(*) > 2;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	count(*)
Electronics	3
Home Appliances	3

Limit to 1000 rows

```
# Business Questions

# Q27. which category generates the most sales volume
select p.category, sum(s.quantity) as total_sales
from sales s
join products p
on s.product_id = p.product_id
group by p.category
order by total_sales desc
limit 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

category	total_sales
Electronics	11

Limit to 1000 rows

```
# Q28. which product sells the most units
select p.product_name, sum(s.quantity) as total_units
from sales s
join products p
on s.product_id = p.product_id
group by p.product_name
order by total_units desc
limit 1;

# Q29. which city has the most customers
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

product_name	total_units
T-Shirt	8

Limit to 1000 rows

```
# Q29. which city has the most customers
select city, count(*) as total_customers
from customers
group by city
order by total_customers desc
limit 1;

# Q30. which product has the highest price
select product_name, max(price)
from products;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

city	total_customers
Delhi	2

Limit to 1000 rows

```
# Q30. which product has the highest price
select product_name, price
from products
order by price desc
limit 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

product_name	price
Laptop	80000.00